Q.26 Describe the working of rack and pinion operated drill jig.

Q.27 What are factors to be considered while designing milling fixture?

Q.28 What are welding fixtures? State the general purpose of a welding fixture.

Q.29 Explain hole basis and shaft basis system.

Q.30 What are the desired properties of a gauge material.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x20=40)

Q.31 Design and draw a suitable drill jig to drill 8 holes of 4mm diameter at 70 mm PCD on 120 mm diameter & 60 mm thick mild steel plate.

Q.32 Design a suitable milling fixture for cutting 8 splines on a shaft of 200 mm length and 50 mm dia.

Q.33 Write the procedure steps for designing a plug gauge with go and not-go side Assuming suitable dimensions draw the same.

(20)

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4th Sem / T&D, CNC, CAD/CAM

Subject:- Jigs Fixtures & Gauges - Design and Drawing

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 A device, which holds, locates a work piece, guides and control one or more cutting tools is called

a) Jig b) Fixture

c) Template d) Lathe

Q.2 Choose the correct statement

a) Fixture is used to guide the tool as well as to locate and clamp the workpiece.

b) Jig is used to guide the tool as well as to locate and clamp the workpiece.

c) Jigs are used on CNC machines to locate and clamp the workpiece and guide the tool

d) No arrangement to guide the tool is provided in jig

Q.3 3-2-1 principle relates to

a) Design of locating devices

b) Work sampling

c) Tool Design

d) Plant layout design

Q.4 Which of the following methods can be used for supporting work on the table?
a) Clamped in a vice
b) Clamped on a block
c) Clamped on the angle plate
d) All of the above

Q.5 Select the operation that does not require a jig
a) Drilling b) Tapping
c) Turning d) Reaming

Q.6 Drilling jig bush is used in Drill jig for
a) Pouring the coolant
b) Locating and guiding the drill
c) Removing the swarf
d) None of the above

Q.7 The following is a quick acting clamp
a) Hinged clamp b) Bridge clamp
c) Cam operated clamp d) Edge clamp

Q.8 The following type of jig is used to drill a series of equidistant hole along a circle
a) Index jig b) Plate type jig
c) Open type jig d) Pot type jig

Q.9 'Go limit' applied to which limit condition?
a) Minimum material limit
b) Lower limit of shaft and upper limit of hole
c) Moderate material limit
d) Maximum material limit

Q.10 What is the use of 'No Go' gauges?
a) Check several dimensions simultaneously
b) Check a single element of a feature
c) Check roundness and size at the same time
d) Check location and size at the same time

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Define limit.
Q.12 Define tolerance.
Q.13 Explain the use of dowel pins.
Q.14 Write the advantage of swing plate clamp.
Q.15 Explain the Function of jig bushes
Q.16 Explain Channel jig.
Q.17 What is a Milling fixture?
Q.18 What are the specific requirements of a turning fixture?
Q.19 Explain the shaft basis system
Q.20 Explain the Bilateral system

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x5=40)

Q.21 Explain interchangeability in mass production. How jigs and fixtures help in achieving it?
Q.22 Differentiate between a jig and a fixture in detail.
Q.23 Explain briefly non-conventional clamping.
Q.24 Explain briefly with neat sketch adjustable locators.
Q.25 Explain briefly box jig and turnover jig.